Journal of Cardiothoracic Surgery

RESEARCH

Open Access

The association between sleep disorders and the risk of atherosclerotic cardiovascular disease: regression analysis and neural network prediction based on NHANES data

Ling-bing Meng[1,2,3†], Kongyong Cui[1,2,3†], Yanjun Song[1,2,3], Bowen Li[1,2,3], Yongbao Zhang[3,4*] and Kefei Dou[1,2,3,4*]

Abstract

Background Sleep quality and duration play a critical role in cardiovascular health. In recent years, sleep disorders have been identified as potential risk factors for atherosclerotic cardiovascular disease (ASCVD). However, their independent effects and underlying mechanisms remain unclear.

Methods This study analyzed data from the National Health and Nutrition Examination Survey (NHANES) conducted between 2013 and 2018, encompassing information on sleep quality, sleep duration, and atherosclerotic cardiovascular disease (ASCVD)-related details from 4,719 participants. Spearman correlation analysis, restricted cubic spline regression, and multivariate logistic regression were employed to evaluate the associations. To enhance predictive modeling under class imbalance, we also developed and compared class-weighted neural networks, random forests, and XGBoost models.

Results The results revealed a significant association between sleep disturbances and ASCVD risk (OR = 1.69, 95% CI: 1.20–2.37, $P = 0.0079$). A "U-shaped" nonlinear relationship was observed between sleep duration and ASCVD, with the lowest risk occurring at 6–8 h of sleep. Compared to 6–8 h of sleep, short sleep (< 6 h) and long sleep (> 8 h) increased ASCVD risk by 45% and 28%, respectively. Among the machine learning models, the neural network demonstrated the highest sensitivity (0.819) and area under the curve (AUC, 0.841) for ASCVD detection, outperforming random forests and XGBoost in identifying true positive cases.

Conclusion In a large, nationally representative U.S. cohort study, self-reported sleep disturbances and abnormal sleep durations were significantly and independently associated with an increased risk of ASCVD. These findings underscore the potential value of sleep health assessment in public health strategies for preventing ASCVD and highlight the utility of neural network models in improving screening sensitivity under conditions of imbalanced data.

[†]Ling-bing Meng and Kongyong Cui contributed equally to this work.

*Correspondence:
Yongbao Zhang
zybzcl@163.com
Kefei Dou
drdoukefei@126.com

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Keywords Sleep disorders, Atherosclerotic cardiovascular disease (ASCVD), NHANES, Neural network

Introduction

Sleep is a vital component of life, playing a crucial role in physiological functions, immune system regulation, and metabolic processes. This regulatory effect not only supports host defense during infections but also maintains inflammatory homeostasis, thereby influencing the development of various chronic diseases [1]. However, with modern lifestyle changes, the prevalence of sleep disorders has significantly increased, becoming a major public health challenge. According to data from the 2017–2020 National Health and Nutrition Examination Survey (NHANES), 29.8% of U.S. adults reported experiencing sleep disorders, and 27.2% experienced daytime sleepiness [2]. Additionally, the 2014 Behavioral Risk Factor Surveillance System (BRFSS) data indicated that over one-third of American adults slept less than seven hours per night, a behavior significantly associated with obesity, diabetes, hypertension, and other health issues [3]. Sleep disorders not only negatively impact individual quality of life and mental health but are also closely linked to multiple chronic diseases, particularly cardiovascular diseases [4–6].

Atherosclerotic cardiovascular disease (ASCVD), a major subtype of cardiovascular diseases, is one of the leading causes of global mortality and disability. Its pathological core lies in atherosclerosis, a process driven by lipid metabolism, chronic inflammation, and endothelial dysfunction [7, 8]. Although traditional risk factors such as elevated low-density lipoprotein cholesterol (LDL-C), hypertension, and smoking have been extensively studied and validated, many patients continue to face residual risks even after significant control of these factors through medications [9–11]. Recently, sleep disorders have emerged as potential risk factors for ASCVD, garnering increasing attention in clinical and epidemiological research [12]. This phenomenon highlights the need to explore less well-defined non-traditional risk factors, among which sleep disorders and abnormal sleep duration may play critical roles.

Abnormal sleep duration may affect cardiovascular health through metabolic, oxidative, and autonomic pathways. Previous studies have shown that sleep disorders may influence cardiovascular health through multiple mechanisms, including activation of the sympathetic nervous system, disruption of circadian rhythms, exacerbation of inflammatory responses, and increased oxidative stress levels [13–16]. Obstructive sleep apnea (OSA) and chronic insomnia have been demonstrated to significantly elevate the risk of cardiovascular diseases [17, 18]. Furthermore, the relationship between sleep duration and cardiovascular metabolic risk exhibits a nonlinear pattern, with both short and long sleep durations associated with increased cardiovascular risk [19]. This "U-shaped" trend highlights that maintaining an optimal sleep duration of 6–8 h may be a crucial strategy for protecting cardiovascular health.

However, most existing studies are limited to specific populations or small sample sizes, lacking large-scale and representative data. Additionally, there remains considerable debate regarding whether sleep disorders serve as independent risk factors for ASCVD and their effects after adjusting for multiple covariates. To address these gaps, this study utilized multi-cycle data from the National Health and Nutrition Examination Survey (NHANES) from 2013 to 2018. Through methods such as restricted cubic spline regression, logistic regression analysis, and neural network modeling, the study systematically examined the relationship between sleep disorders, sleep duration, and ASCVD risk.

The objectives of this study were to elucidate the potential mechanisms by which sleep-related factors influence the risk of ASCVD, verify the role of sleep disorders as independent risk factors for ASCVD, and explore precise prevention strategies for ASCVD based on sleep health management. This study explores the link between sleep problems and ASCVD to guide prevention and integrate sleep management into public health.

Materials and methods

Clinical data

The data for this study were obtained from the National Health and Nutrition Examination Survey (NHANES) (https://www.cdc.gov/nchs/nhanes/index.htm), a cross-sectional survey conducted by the National Center for Health Statistics (NCHS) in collaboration with the Centers for Disease Control and Prevention (CDC). NHANES aims to provide nationally representative data on the civilian, non-institutionalized population of the United States.

The data collection protocol was approved by the NCHS Ethics Review Board, and all survey participants provided informed consent before undergoing interviews and examinations. In this study, data from three NHANES cycles (2013–2014, 2015–2016, and 2017–2018) were combined into a dataset using publicly available response data files. The study population included all respondents from these NHANES cycles.

Data collection

The covariates used in this study were selected based on prior literature and substantive reasoning and were considered to have potential associations with sleep quality and cardiovascular disease prevalence. Information on a

variety of factors was collected using standardized questionnaires, including age, gender, race, education level, marital status, poverty-to-income ratio (PIR), body mass index (BMI), sedentary time, moderate activity duration, smoking status, alcohol use, and medical comorbidities such as hypertension, hyperlipidemia, and diabetes. Missing data were supplemented using imputation methods implemented in R software.

Survey data from 17,962 participants were initially collected for this study. After excluding individuals with missing outcome or exposure data, as well as those who did not provide complete information on BMI, education level, marital status, PIR, smoking, or alcohol use, a total of 4,719 participants were included in the final analysis (See Table 1 for details).

According to the 2013 American College of Cardiology/American Heart Association (ACC/AHA) Guideline on the Treatment of Blood Cholesterol to Reduce Atherosclerotic Cardiovascular Risk in Adults, ASCVD is defined as having at least one diagnosis of coronary heart

Table 1 Baseline Characteristics

Variable		ASCVD		*P*-value
		No (*n* = 4360)	Yes (*n* = 35)	
Age		46 (33 ~ 60)	67 (57 ~ 75)	< 0.001
Gender	Male	2039 (46.77%)	191 (53.20%)	0.019
	Female	2321 (53.23%)	168 (46.80%)	
Race	Mexican American	567 (13.00%)	35 (9.75%)	0.203
	Non-Hispanic White	377 (8.65%)	29 (8.08%)	
	Non-Hispanic Black	1883 (43.19%)	197 (54.87%)	
	Non-Hispanic Asian	872 (20.00%)	70 (19.50%)	
	Other Hispanic	661 (15.16%)	28 (7.80%)	
Education level	< high school	299 (6.86%)	43 (11.98%)	< 0.001
	nine-eleven year	552 (12.66%)	63 (17.55%)	
	high school	964 (22.11%)	95 (26.46%)	
	college or above	2545 (58.37%)	158 (44.01%)	
Marial status	live alone	2606 (59.77%)	204 (56.82%)	0.274
	not live alone	1754 (40.23%)	155 (43.18%)	
Income status	low income	1475 (33.83%)	156 (43.45%)	< 0.001
	middle income	1486 (34.08%)	132 (36.77%)	
	high income	1399 (32.09%)	71 (19.78%)	
AIP		1.69 (1.56,1.8)	1.82 (1.53,2.1)	0.049
BMI		29.07 ± 7.13	30.58 ± 8.04	< 0.001
Sitting time(min)		441.05 ± 520.14	447.28 ± 202.19	0.822
Physical activity time(min)		63.04 ± 58.65	66.02 ± 98.21	0.619
Sleep duration (hours)	< 7	1686 (38.67%)	165 (45.96%)	0.151
	7–9	2551 (58.51%)	173 (48.19%)	
	> 9	123 (2.82%)	21 (5.85%)	
Sleep disorder	No	3154 (72.34%)	167 (46.52%)	< 0.001
	Yes	1206 (27.66%)	192 (53.48%)	
Smoke	Never	2535 (58.14%)	149 (41.50%)	< 0.001
	Former	939 (21.54%)	122 (33.98%)	
	Current	886 (20.32%)	88 (24.51%)	
Drink	Never Drank	1537 (35.25%)	184 (51.25%)	< 0.001
	Occasionally Drank	1519 (34.84%)	114 (31.75%)	
	Regularly Drank	955 (21.90%)	38 (10.58%)	
	Heavy Drinker	349 (8.00%)	23 (6.41%)	
Hypertension	No	3800 (87.16%)	199 (55.43%)	< 0.001
	Yes	560 (12.84%)	160 (44.57%)	
Hyperlipidemia	No	2700 (61.93%)	240 (66.85%)	0.064
	Yes	1660 (38.07%)	119 (33.15%)	
Diabetes	No	3877 (88.92%)	225 (62.67%)	< 0.001
	Yes	483 (11.08%)	134 (37.33%)	

BMI, body mass index; PIR, poverty income ratio; AIP, Atherogenic Index of Plasma; ASCVD, atherosclerotic cardiovascular disease

disease, angina, myocardial infarction, or stroke, with myocardial infarction and stroke classified as strict criteria. In this study, the assessment of ASCVD was based on a series of self-reported questionnaires. Participants were identified as having ASCVD if they responded affirmatively to the NHANES multiple-choice question (MCQ): "Have you ever been told you have coronary heart disease, angina, a heart attack, or a stroke?"

Sleep disorders were identified based on affirmative responses to either of the following questions: "Have you ever told a doctor or other health professional that you have trouble sleeping?" or "Has a doctor or other health professional ever told you that you have a sleep disorder?" Sleep duration was self-reported in response to the question: "How many hours of sleep do you usually get at night on weekdays or workdays?" The reported sleep duration was categorized as short (< 7 h/night), normal (7–9 h/night), and long (> 9 h/night).

Hypertension was defined as participants self-reporting a diagnosis of hypertension, having an average systolic blood pressure (SBP) ≥ 130 mmHg or an average diastolic blood pressure (DBP) ≥ 80 mmHg, or currently taking antihypertensive medications. Hyperlipidemia was defined based on the criteria of the National Cholesterol Education Program (NCEP) Adult Treatment Panel III (ATP III), including total cholesterol ≥ 200 mg/dL, triglycerides ≥ 150 mg/dL, high-density lipoprotein cholesterol (HDL-C) levels < 40 mg/dL for men or < 50 mg/dL for women, or low-density lipoprotein cholesterol (LDL-C) levels ≥ 130 mg/dL. Additionally, participants who reported using lipid-lowering medications were also classified as having hyperlipidemia.

Statistical analysis

Statistical analyses were performed using SPSS 25.0 software, and the sample was divided into two groups: ASCVD and non-ASCVD. Continuous variables were presented as mean ± standard deviation (SD), while categorical variables were expressed as numbers and percentages. For normally distributed continuous data, the results were described as mean ± SD and evaluated using t-tests to assess group differences. For non-normally distributed data, results were described as medians with interquartile ranges (M [P25, P75]).

First, Spearman correlation analysis was conducted to explore the association between ASCVD and the included variables. Then, multiple linear regression was used to analyze the influencing factors of the atherosclerotic index. The nonlinear relationship between sleep duration and ASCVD was explored by applying restricted cubic spline (RCS) regression. Multivariate logistic regression was used to evaluate the association between sleep disorders and ASCVD, and demographic, lifestyle and clinical variables were adjusted in sequence. Finally, machine learning predictive modeling was adopted, incorporating 13 variables including age, gender, BMI, sleep disorders, sleep duration, education, income, smoking, drinking, hypertension, diabetes, hyperlipidemia, and sedentary time. Compare three algorithms: neural network, random forest and XGBoost. The training set (80%) and the test set (20%) were hierarchically divided according to the ASCVD status, and category weighting was adopted to handle sample imbalance. Calculate the accuracy rate, precision rate, recall rate, F1 score and AUC-ROC in the test set. The AUC of the models was compared using Bootstrap resampling (1000 times) and corrected by Bonferroni. The statistical significance was set at $P \le 0.05$.

Results

Baseline characteristics of patients

The distribution of patient characteristics grouped by the presence or absence of ASCVD revealed significant differences across multiple clinical variables in the preliminary analysis.

Among individuals with ASCVD ($n = 359$) and those without ASCVD ($n = 4360$), there were notable disparities. Specifically, in the group with sleep disorders, 192 individuals (53.48%) had ASCVD, compared to 167 individuals (46.52%) in the group without sleep disorders. This indicates that sleep disorders are significantly associated with an increased risk of ASCVD ($P < 0.001$).

The mean age of the ASCVD group was 67 years (range: 57–75), significantly higher than the mean age of 46 years (range: 33–60) in the non-ASCVD group ($P < 0.001$). Regarding educational level and household income, a higher proportion of individuals with ASCVD had lower income and education levels ($P < 0.001$), suggesting that lower socioeconomic status may increase the risk of ASCVD.

The median Atherogenic Index of Plasma (AIP) in the ASCVD group was 1.82 (interquartile range: 1.53–2.18), higher than the 1.69 (1.56–1.88) observed in the non-ASCVD group ($P = 0.049$). The mean BMI of the ASCVD group was 30.58 ± 8.04, significantly higher than 29.07 ± 7.13 in the non-ASCVD group ($P < 0.001$).

In terms of chronic diseases, the prevalence of diabetes and hypertension in the ASCVD group was slightly lower than that in the non-ASCVD group ($P < 0.001$). There was a significant difference in the prevalence of hyperlipidemia ($P = 0.064$).

Additionally, the ASCVD group had a higher proportion of current or former smokers and a lower proportion of individuals who had never consumed alcohol compared to the non-ASCVD group ($P < 0.001$) (Table 1).

Table 2 Spearman correlation analysis of ASCVD

Variable	Correlation Coefficient	*P*-value
Age	0.272**	< 0.01
Gender	-0.034*	0.019
Race	-0.0024	0.106
BMI	0.051**	< 0.01
Marital status	0.016	0.274
Education level	0.082**	< 0.01
Income level	-0.072**	< 0.01
Sitting time	0.035*	0.017
physical active time	-0.043	0.058
Smoke	0.077**	< 0.01
Drink	-0.065	0.233
Hypertension	0.234**	< 0.01
Hyperlipidemia	0.027	0.064
Diabetes	0.206**	< 0.01
Sleep disorder	0.150**	< 0.01
Sleep quality status	0.055**	< 0.01

*When the confidence level (double test) is 0.05, the correlation is significant

**When the confidence level (double test) is 0.01, the correlation is significant

Correlation analysis of ASCVD

Sleep duration was converted into a binary sleep quality variable: abnormal sleep duration (> 9 h/night or < 7 h/night) was coded as 1, while normal sleep duration (7–9 h/night) was coded as 0. According to the results of Spearman correlation analysis, sleep disorders, sleep quality, diabetes, hypertension, smoking, sedentary behavior, education level, income level, gender, age, and BMI were significantly correlated with ASCVD.

Specifically, the presence of sleep disorders and abnormal sleep quality were positively associated with ASCVD ($r = 0.150^{**}$, $p < 0.01$; $r = 0.055^{**}$, $p < 0.01$), indicating that individuals with poor sleep quality may have a higher likelihood of developing ASCVD. Additionally, ASCVD risk increased significantly with age ($r = 0.272^{**}$, $p < 0.01$). Smoking, sedentary behavior, hypertension, and diabetes also showed significant positive correlations with ASCVD ($p < 0.01$). Conversely, income level was negatively correlated with ASCVD ($p < 0.01$), suggesting that socioeconomic status influences the prevalence of cardiovascular disease (Table 2).

Exploration of the Nonlinear Relationship Between Sleep Duration and ASCVD Using Restricted Cubic Spline (RCS) Regression

In the RCS regression analysis, a significant nonlinear relationship between sleep duration and ASCVD was detected after adjusting for potential covariates (P for nonlinearity < 0.001). The relationship exhibited a pronounced 'U-shaped' pattern, with the lowest risk observed at approximately 6 to 8 h of sleep per night. Sleep durations under 6 or over 8 h were associated with progressively higher ASCVD risk (Fig. 1). As illustrated in Fig. 1, the confidence bands widened at the extremes of sleep duration (particularly < 4 h and > 9 h), reflecting reduced precision in risk estimates due to sparse data in these ranges. This pattern underscores that both short and long sleep durations may confer increased ASCVD risk, with optimal protection observed within the 6–8 h range (Figs. 1 and 2).

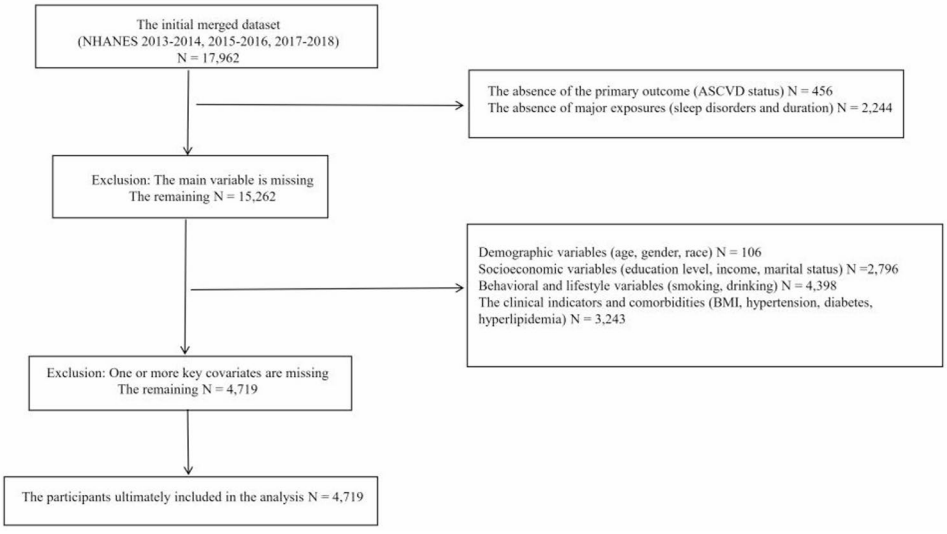


Fig. 1 Flowchart of Study Participant Selection. This study was based on data from the National Health and Nutrition Examination Survey (NHANES) 2013–2018. The initial merged dataset comprised 17,392 participants. Exclusion criteria were applied sequentially: first, participants with missing primary variables (ASCVD status, sleep disorders, and sleep duration; $n = 456 + 2,244$) were excluded; second, participants missing one or more key covariates (including demographic, socioeconomic, behavioral, and clinical indicators) were excluded (total excluded = 10,873). Consequently, a total of 4,719 participants were included in the final analysis

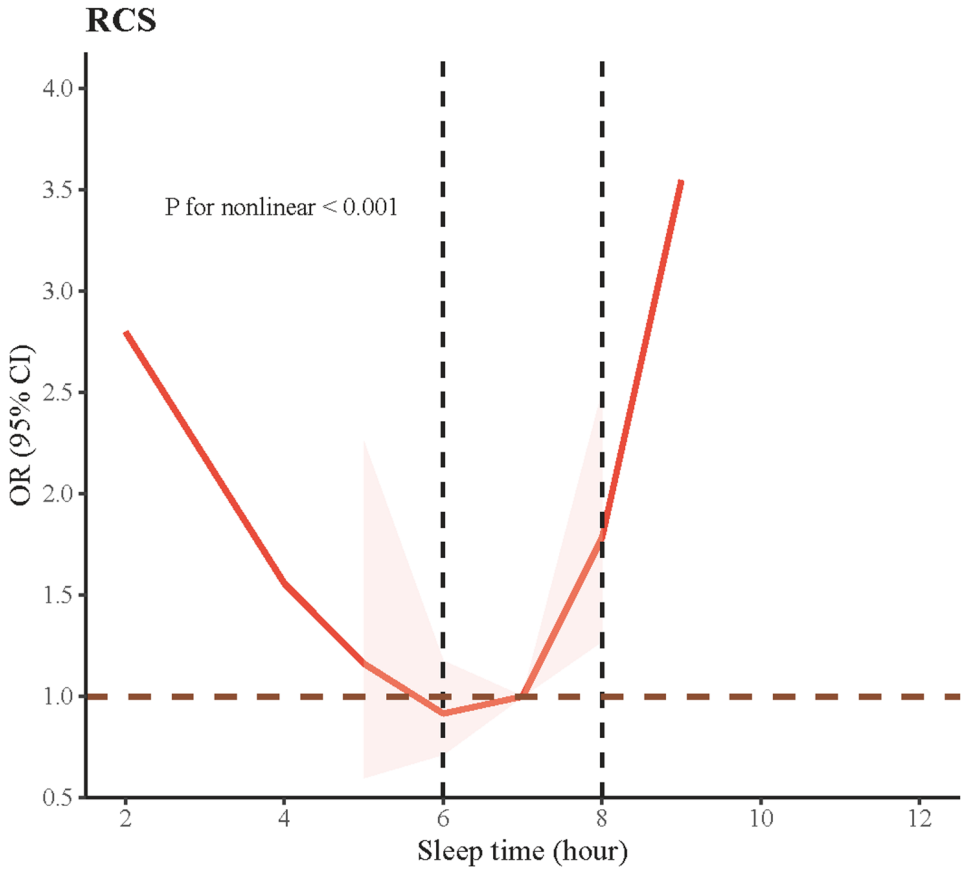


Fig. 2 Determination of the association between Sleep time(hour) and ASCVD by restricted cubic spline (RCS) regression analysis. The chart demonstrates a significant nonlinear relationship between sleep time(hour) and the odds ratio (OR) for atherosclerotic cardiovascular disease. As the sleep time increases, the OR shows "U" shape association, indicating that this relationship is statistically significant ($p < 0.001$). The shaded area represents the 95% confidence interval for the OR estimate, illustrating the uncertainty around the OR estimate

Multivariate linear regression analysis of factors associated with the Aherogenic Index of Plasma (AIP)

The Atherogenic Index of Plasma (AIP) is an important biomarker associated with ASCVD risk. To identify factors independently correlated with AIP levels, a multivariate linear regression model was constructed with AIP as the dependent variable. The results indicated that ASCVD status was significantly associated with higher AIP levels ($\beta = 0.134$, $P = 0.041$). Additionally, older age ($\beta = 0.004$, $P = 0.033$), race ($\beta = 0.040$, $P = 0.036$), and the presence of hyperlipidemia ($\beta = 0.271$, $P < 0.001$) were also independently associated with elevated AIP levels (Table 3). This analysis delineates the factors that correlate with this atherogenic lipid marker, rather than implying a causal or mediational pathway to ASCVD.

Logistic Regression Analysis of the Association Between Sleep Disorders and ASCVD

Multivariable logistic regression analysis was conducted to determine whether sleep disorders are a risk factor for ASCVD, using participants without sleep disorders as the reference group. In the unadjusted analysis, the odds ratio (OR) for ASCVD risk associated with sleep disorders was 2.04 (95% CI: 1.44–2.89; $P < 0.001$).

After adjusting for age, gender, and race (Model 1), the OR decreased to 1.77 (95% CI: 1.22–2.57; $P = 0.0029$). Further adjustment for BMI categories, sedentary time, and physical activity duration (Model 2) slightly reduced the OR to 1.73 (95% CI: 1.19–2.52; $P = 0.0041$). In the fully adjusted model (Model 3), which accounted for marital status, educational level, income level, smoking, and alcohol consumption, the OR for ASCVD risk associated with sleep disorders was 1.69 (95% CI: 1.15–2.49; $P = 0.0079$).

The results demonstrated high statistical significance, and although the OR values varied slightly across models, the association between sleep disorders and ASCVD remained significant after adjusting for all covariates. This indicates that sleep disorders are an independent risk factor for ASCVD. These findings underscore the importance of improving sleep quality and addressing sleep disorders as key strategies for ASCVD prevention (Table 4).

Table 3 Multivariate linear regression analysis of factors associated with the Atherogenic Index of Plasma (AIP)

-	B	S. E	t	*P*-value	VIF
Age	0.004	0.002	2.145	0.033	1.181
Gender	0.050	0.047	1.050	0.295	1.217
Race	0.040	0.019	2.109	0.036	1.183
Marital status	0.065	0.047	1.388	0.166	1.207
Education level	-0.003	0.025	-0.141	0.888	1.344
Income level	-0.018	0.048	-0.374	0.709	1.255
Sitting time	0.000	0.000	-0.391	0.696	1.103
Physical active time	0.000	0.000	-1.006	0.315	1.078
Smoke	0.020	0.031	0.652	0.515	1.298
Drink	0.022	0.025	0.887	0.376	1.154
Hypertension	0.067	0.062	1.090	0.277	1.088
Hyperlipidemia	0.271***	0.045	6.083	< 0.001	1.082
ASCVD	0.134**	0.065	2.051	0.041	1.138

*When the confidence level (double test) is 0.05, the correlation is significant

**When the confidence level (double test) is 0.01, the correlation is significant

Table 4 Logistic regression analysis of the association between sleep disorder and ASCVD

-	B	S. E	OR (95%CI)	*P*-value
Un adjusted	0.714	0.177	2.04(1.44,2.89)	< 0.001
Model 1	0.570	0.191	1.77(1.22,2.57)	0.0029
Model 2	0.550	0.192	1.73(1.19,2.52)	0.0041
Model 3	0.524	0.197	1.69(1.15,2.49)	0.0079

Model 1 was adjusted for Age, Gender, Race

Model 2 was adjusted for Age, Gender, Race, BMI group, Sitting time, Physical active time

Model 3 was adjusted for Age, Gender, Race, BMI group, Marital status, Education level, Income level, Sitting time, Physical active time, Smoke, Drink

Table 5 Performance comparison of three machine learning algorithms on the test set ($n = 944$)

Algorithm	Accuracy	Precision	Recall (sensitivity)	F1 score	AUC-ROC
Neural network	0.733	0.198	0.819	0.319	0.841
Random Forest	0.925	0.600	0.042	0.078	0.833
XGBoost	0.775	0.218	0.750	0.337	0.839

Comparison of the predictive performance of machine learning models for ASCVD

The neural network model performs outstandingly

Based on a unified dataset ($N = 4{,}719$), we constructed three models: neural network, random forest, and XGBoost for comparison. All models adopted the same training set ($n = 3{,}775$, 80%) and test set ($n = 944$, 20%), among which the positive proportion of ASCVD in the test set was 7.6% (72/944).

As shown in Table 5, the neural network performed the best in identifying ASCVD positive cases, with a recall rate (sensitivity) as high as 0.819, and the AUC-ROC value also reached the highest of 0.841 among the three models. This indicates that neural networks can capture the complex nonlinear interactions among sleep disorders, metabolic indicators and traditional risk factors, thereby more effectively identifying high-risk individuals. In contrast, although XGBoost also demonstrated a relatively high recall rate (0.750), it was slightly lower than that of neural networks. Random forests exhibit a conservative prediction model with a high precision rate (0.600) but an extremely low recall rate (0.042).

Statistical comparison shows that the discriminative abilities of the three models are comparable

To rigorously assess the statistical significance of AUC differences between models, we conducted pairwise comparisons using Bootstrap resampling (1,000 times) and applied Bonferroni correction to control multiple comparison errors (corrected significance level $\alpha = 0.0167$). The results are shown in Table 6.

None of the comparisons reached the corrected statistical significance threshold, indicating that the three models are comparable in overall discriminative ability (AUC). It is worth noting that in the direct comparison with neural networks, the AUC differences were all negative (-0.0075 and −0.0021), indicating that the AUC of neural networks maintained a stable leading position in numerical terms. This finding highlights the distinction between numerical trends and statistical significance: although the neural network consistently performed

Table 6 Statistical comparison of AUC values among algorithms

Comparison	AUC difference	95% confidence interval	*P*-value
Random Forest vs. Neural Network	-0.0075	(-0.0351, 0.0197)	0.646
XGBoost vs. Neural Network	-0.0021	(-0.0227, 0.0183)	0.842
Random Forest vs. XGBoost	-0.0054	(-0.0304, 0.0203)	0.672

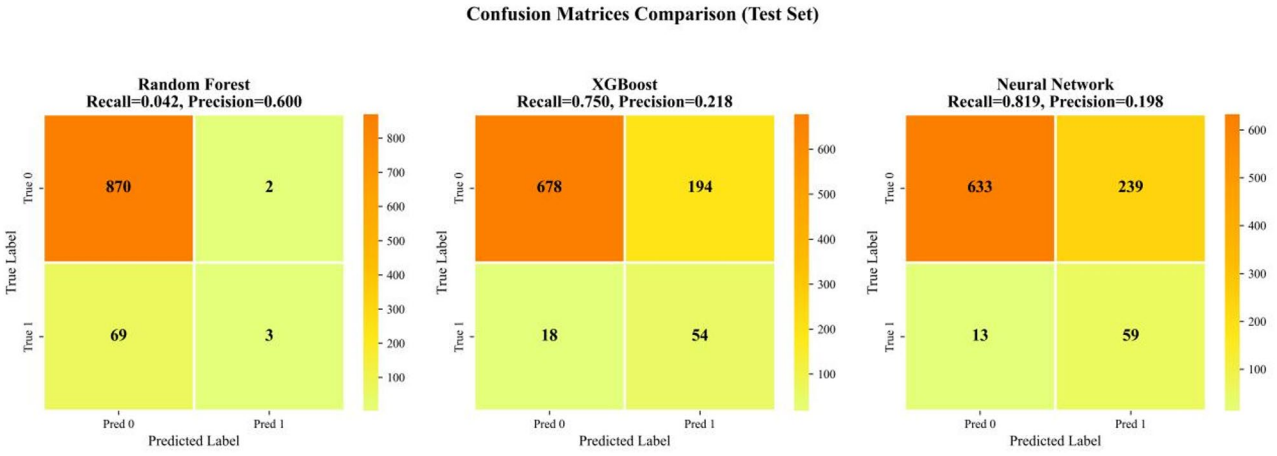


Fig. 3 Confusion matrix comparison chart. This figure presents side-by-side heatmaps of the confusion matrices for the Random Forest, XGBoost, and Neural Network models on the independent test set ($n = 944$). The X-axis represents the Predicted Label ("0" for non-ASCVD and "1" for ASCVD). The Y-axis represents the True Label ("0" for non-ASCVD and "1" for ASCVD). The number in each cell indicates the count of samples falling into that specific combination, and the color intensity visually represents the magnitude of the count (darker color typically indicates a higher count). The recall (sensitivity) and precision for the ASCVD-positive class (label 1) are annotated for each model. The results show that the Neural Network achieved the highest sensitivity (Recall = 0.819) in identifying ASCVD cases but had a relatively low precision (Precision = 0.198). In contrast, the Random Forest exhibited the highest precision (Precision = 0.600) but very low sensitivity (Recall = 0.042)

best in the AUC metric (0.841) and was not surpassed by other models in pairwise comparisons, the three algorithms demonstrated considerable discriminative efficacy in a statistical sense. Therefore, when choosing a model, more consideration can be given to the numerical advantages of neural networks and their high recall rate characteristics, especially in application scenarios with high requirements for screening sensitivity.

The unique advantages of neural networks in screening scenarios

The confusion matrix comparison chart (Fig. 3) clearly reveals the predictive advantages of the neural network: among the 72 ASCVD positive samples in the test set, the neural network correctly identified 59 cases (true positives) and only missed 13 cases, significantly outperforming the other two models. This high recall rate feature (81.9%) makes it particularly suitable as a primary screening tool, minimizing the risk of missed diagnosis and conforming to the principle of "it is better to misreport than to miss" in public health screening. Although the neural network generated a relatively large number of false positives (240 cases), this is an acceptable trade-off in the screening scenario - which can be reconfirmed through more specific tests (such as random forest models or clinical evaluations) later.

The ROC curve comparison graph (Fig. 4) visually demonstrates that all three models have good discriminative ability (AUC > 0.83), and the curves are generally close, further supporting their comparability in discriminative efficacy.

The neural network model constructed in this study not only verified the statistical significance of sleep disorders as an independent risk factor for ASCVD, but also demonstrated its unique value in early screening and risk stratification through its outstanding high recall rate. In application scenarios that require high sensitivity (such as initial screening of large-scale populations), neural networks are a more suitable choice. Future research can construct a more balanced prediction system by combining the advantages of different algorithms through methods such as ensemble learning.

Discussion

ASCVD is a leading cause of global mortality and disability, underscoring its significant public health implications [20]. Although the effects of hypertension, dyslipidemia and smoking have been fully confirmed, this study contributes to the growing evidence that sleep disorders are associated with the risk of ASCVD [12]. By analyzing nationally representative NHANES data, we systematically demonstrated a significant association between

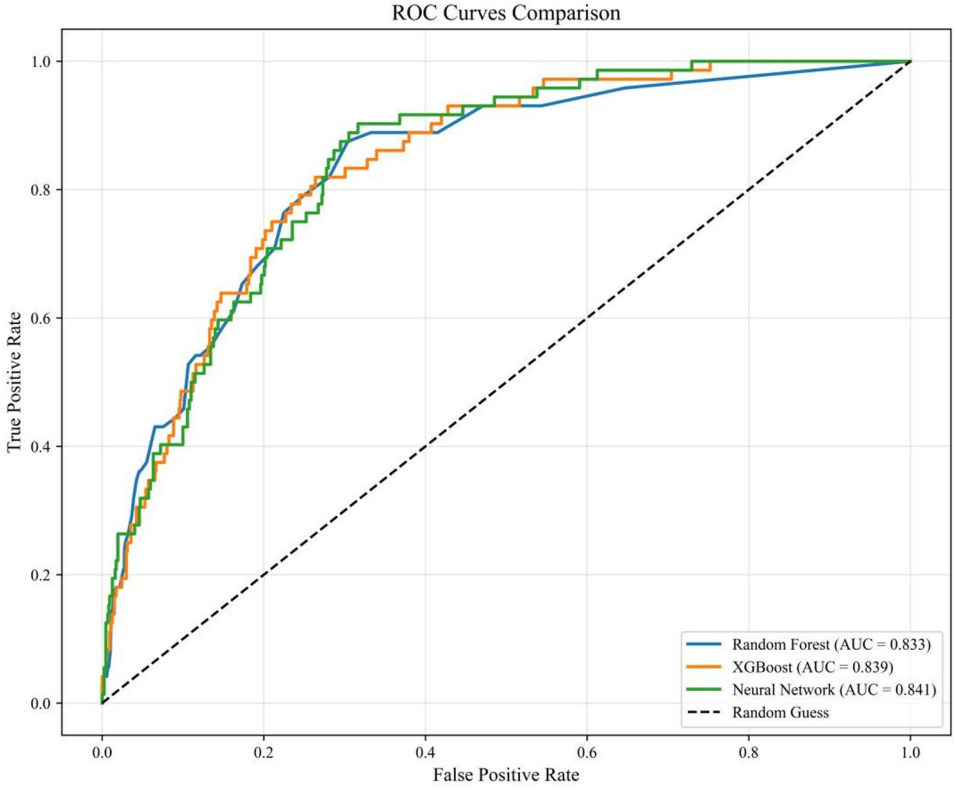


Fig. 4 ROC curve comparison chart. This figure displays the Receiver Operating Characteristic (ROC) curves for the Random Forest, XGBoost, and Neural Network models. The X-axis is the False Positive Rate (1 - Specificity), representing the proportion of actual negative cases incorrectly predicted as positive. The Y-axis is the True Positive Rate (Sensitivity/Recall), representing the proportion of actual positive cases correctly identified. The diagonal dashed line indicates the performance of a random guess model (AUC = 0.5). A curve closer to the top-left corner indicates better overall discriminatory performance. The area under the curve (AUC) values are annotated: Neural Network (AUC = 0.841), XGBoost (AUC = 0.839), and Random Forest (AUC = 0.833). All models performed significantly better than random chance, with comparable AUC values among them, and the Neural Network showed a slight advantage

self-reported sleep disorders, abnormal sleep duration and ASCVD. Our findings are consistent with previous studies and expand existing research by using robust statistical methods to simulate complex relationships and explore the application of machine learning in this predictive context.

In this study, restricted cubic spline regression analysis revealed a "U-shaped" nonlinear relationship between sleep duration and ASCVD risk. Specifically, individuals sleeping 6 to 8 h per night exhibited the lowest ASCVD risk, whereas those sleeping less than 6 h or more than 8 h had significantly increased risks. This finding aligns with previous research, further confirming the importance of sleep duration for cardiovascular health [21].

A Mendelian randomization study further validated this "U-shaped" nonlinear relationship, showing that genetically predicted short sleep duration (≤ 6 h) significantly increased the risk of coronary artery disease, myocardial infarction, and arterial hypertension, while long sleep duration (≥ 9 h) was not associated with a significant causal relationship [22]. Additionally, prior studies have suggested that short sleep duration may elevate ASCVD risk through various mechanisms [23]. The meta-analysis by Kwok et al., which included objective sleep measurement data, also supported the "U-shaped" dose-response relationship [24]. On the other hand, sleep duration may also be closely linked to conditions such as depression, chronic inflammation, and cardiac dysfunction [14, 25, 26].

Logistic regression analysis indicated that sleep disorders were independently associated with the risk of ASCVD. After comprehensive adjustments for covariates including age, gender, BMI, income level, and health behaviors, the significant association between sleep disorders and ASCVD risk persisted. For instance, Marin et al. 's research based on polysomnography shows that severe sleep apnea is significantly associated with the risk of cardiovascular events [27]. Sleep disorders may impact cardiovascular health via metabolic, oxidative, and autonomic pathways, driving atherosclerosis, which contribute to the development and progression of atherosclerosis. These mechanisms are consistent with previous literature discussing the interaction between sleep disorders, oxidative stress, and inflammatory responses [15, 28, 29].

Moreover, the study found that the prevalence of sleep disorders was significantly higher among middle-aged and older adults, particularly those with other chronic conditions such as hypertension and diabetes [19]. This finding underscores the importance of prioritizing sleep health in public health interventions targeting middle-aged and older populations to reduce the risk of ASCVD.

Through multiple linear regression analysis, this study further identified independent factors related to the plasma atherosclerotic index (AIP), including age, hyperlipidemia, and racial differences, etc. This analysis aims to identify the variables related to AIP levels rather than infer their mediating role in the association between sleep disorders and ASCVD. As a potential biomarker reflecting lipid metabolism and the risk of atherosclerosis, the study of the influencing factors of AIP is helpful for a more comprehensive understanding of the risk composition of ASCVD in clinical assessment. Future studies can further explore the potential mechanism role of AIP in sleep-cardiovascular associations in longitudinal designs.

In addition, this study attempts to construct a machine learning model for risk prediction. Among them, the neural network model demonstrated high sensitivity on the test set (with a recall rate of 0.819), and has potential application value in the initial screening. Although its overall discriminative ability is comparable to that of other models, its high sensitivity is more in line with the principle of "it is better to give a false alarm than to miss a diagnosis" in public health screening.

This study also has several limitations. Firstly, cross-sectional design cannot infer causal relationships. Secondly, both sleep disorders and ASCVD are based on self-reporting, which may involve recall bias and classification bias, and do not distinguish specific types of sleep disorders (such as obstructive sleep apnea and insomnia). Thirdly, the large amount of missing data in the sample may introduce selection bias. Methodological studies have shown that abnormal data in health surveys are associated with poorer health status [30], which implies that excluded individuals may have a higher baseline risk, leading to a conservative estimation of association strength in this study. Finally, the research sample was derived from the American population, and caution should be exercised when extrapolating the conclusion to other populations.

Nevertheless, the results of this study still have public health significance. It suggests that sleep assessment should be incorporated into the primary prevention system of cardiovascular diseases, and more attention should be paid to sleep problems in clinical practice. Future research should combine prospective design, objective sleep monitoring tools (such as polysomnography and body movement recorders), and clinically determined endpoint events to further clarify the longitudinal association between sleep and ASCVD, and distinguish the impact of different types of sleep disorders.

Conclusion

In summary, this study utilized multi-period NHANES data and found that self-reported sleep disorders and abnormal sleep duration were significantly associated with the risk of ASCVD, further supporting the "U-shaped" relationship between sleep duration and ASCVD. Studies have shown that sleep problems evaluated through questionnaires at the population level can serve as an independent and relevant indicator of ASCVD risk. These findings provide a basis for incorporating sleep health management into the prevention of cardiovascular diseases and also offer a reference for using machine learning for early risk identification. In the future, research that combines objective measurement with prospective design is still needed to further clarify its clinical and public health value.

Abbreviations

NHANES	National Health and Nutrition Examination Survey
BRFSS	Behavioral Risk Factor Surveillance System
ASCVD	Atherosclerotic cardiovascular disease
LDL-C	Low-density lipoprotein cholesterol
OSA	Obstructive sleep apnea
NCHS	The National Center for Health Statistics
CDC	The Centers for Disease Control and Prevention
PIR	Poverty-to-income ratio
BMI	Body mass index
MCQ	Multiple-choice questions
PHQ-9	The Patient Health Questionnaire

Acknowledgements

Not applicable.

Author contributions

Ling-bing Meng make the acquisition, analysis, Kongyong Cui interpretate of data; Yanjun Song and Bowen Li create of new software used in the work; Yongbao Zhang and Kefei Dou have drafted the work or substantively revised it.

Funding

The research was funded by National Natural Science Foundation of China (Grant no. 82400543, 82304864), CAMS Innovation Fund for Medical Sciences (CIFMS) (2025-12M-XHXX-034), Noncommunicable ChronicDiseases-National Science and Technology Major Project, 2025ZD0548200, The Noncommunicable Chronic Diseases-National Science and Technology Major Project (Grant Number: 2023ZD0513900), the Fundamental Research Funds for the Central Universities (Grant Number: 3332024033).

Data availability

The research data was sourced from the NHANES database, with the data website being https://www.cdc.gov/nchs/nhanes/index.html.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Cardiometabolic Medicine Center, National Clinical Research Center for Cardiovascular Diseases, Fuwai Hospital, National Center for Cardiovascular Diseases, Chinese Academy of Medical Sciences and Peking Union Medical College, A 167 Beilishi Rd, Xicheng District, Beijing 100037, China
[2]Department of Cardiology, Fuwai Hospital, National Center for Cardiovascular Diseases, National Clinical Research Center for Cardiovascular Diseases, Chinese Academy of Medical Sciences and Peking Union Medical College, A 167 Beilishi Rd, Xicheng District, Beijing 100037, China
[3]State Key Laboratory of Cardiovascular Disease, A 167 Beilishi Rd, Xicheng District, Beijing 100037, China
[4]Aortic and Vascular Surgery Centre, CAMS & PUMC, Fuwai Hospital, National Centre for Cardiovascular Diseases, Beijing, China

Received: 10 July 2025 / Accepted: 23 February 2026
Published online: 02 March 2026

References

1. Besedovsky L, Lange T, Haack M. The Sleep-Immune Crosstalk in Health and Disease. Physiol Rev. 2019;99(3):1325–80.
2. Di H, Guo Y, Daghlas I, et al. Evaluation of Sleep Habits and Disturbances Among US Adults, 2017–2020. JAMA Netw Open. 2022;5(11):e2240788.
3. Liu Y, Wheaton AG, Chapman DP, Cunningham TJ, Lu H, Croft JB. Prevalence of Healthy Sleep Duration among Adults–United States, 2014. MMWR Morb Mortal Wkly Rep. 2016;65(6):137–41.
4. Reimer MA, Flemons WW. Quality of life in sleep disorders. Sleep Med Rev. 2003;7(4):335–49.
5. Lyons OD. Sleep disorders in chronic kidney disease. Nat Rev Nephrol. 2024;20(10):690–700.
6. Covassin N, Somers VK. Sleep, melatonin, and cardiovascular disease. Lancet Neurol. 2023;22(11):979–81.
7. Sandesara PB, Virani SS, Fazio S, Shapiro MD. The Forgotten Lipids: Triglycerides, Remnant Cholesterol, and Atherosclerotic Cardiovascular Disease Risk. Endocr Rev. 2019;40(2):537–57.
8. Lawler PR, Bhatt DL, Godoy LC, et al. Targeting cardiovascular inflammation: next steps in clinical translation. Eur Heart J. 2021;42(1):113–31.
9. Catapano AL, Graham I, De Backer G, et al. 2016 ESC/EAS Guidelines for the Management of Dyslipidaemias. Eur Heart J. 2016;37(39):2999–3058.
10. Stone NJ, Robinson JG, Lichtenstein AH, et al. 2013 ACC/AHA guideline on the treatment of blood cholesterol to reduce atherosclerotic cardiovascular risk in adults: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol. 2014;63(25 Pt B):2889–934.
11. Ference BA, Braunwald E, Catapano AL. The LDL cumulative exposure hypothesis: evidence and practical applications. Nat Rev Cardiol. 2024;21(10):701–16.
12. Libby P. The changing landscape of atherosclerosis. Nature. 2021;592(7855):524–33.
13. Spiegel K, Leproult R, Van Cauter E. Impact of sleep debt on metabolic and endocrine function. Lancet. 1999;354(9188):1435–9.
14. Irwin MR. Sleep and inflammation: partners in sickness and in health. Nat Rev Immunol. 2019;19(11):702–15.
15. Vaccaro A, Kaplan Dor Y, Nambara K, et al. Sleep Loss Can Cause Death through Accumulation of Reactive Oxygen Species in the Gut. Cell. 2020;181(6):1307–e132815.
16. Chellappa SL, Vujovic N, Williams JS, Scheer F. Impact of Circadian Disruption on Cardiovascular Function and Disease. Trends Endocrinol Metab. 2019;30(10):767–79.
17. Redline S, Azarbarzin A, Peker Y. Obstructive sleep apnoea heterogeneity and cardiovascular disease. Nat Rev Cardiol. 2023;20(8):560–73.
18. Thurston RC, Chang Y, Kline CE, et al. Trajectories of Sleep Over Midlife and Incident Cardiovascular Disease Events in the Study of Women's Health Across the Nation. Circulation. 2024;149(7):545–55.
19. St-Onge MP, Grandner MA, Brown D, et al. Sleep Duration and Quality: Impact on Lifestyle Behaviors and Cardiometabolic Health: A Scientific Statement From the American Heart Association. Circulation. 2016;134(18):e367–86.
20. Nayor M, Brown KJ, Vasan RS. The Molecular Basis of Predicting Atherosclerotic Cardiovascular Disease Risk. Circulation ResearchCirculation ResearchCirculation Res. 2021;128(2):287–303.
21. Linz D, Kadhim K, Kalman JM, McEvoy RD, Sanders P. Sleep and cardiovascular risk: how much is too much of a good thing. Eur Heart J. 2019;40(20):1630–2.
22. Ai S, Zhang J, Zhao G, et al. Causal associations of short and long sleep durations with 12 cardiovascular diseases: linear and nonlinear Mendelian randomization analyses in UK Biobank. Eur Heart J. 2021;42(34):3349–57.
23. Cappuccio FP, Cooper D, D'Elia L, Strazzullo P, Miller MA. Sleep duration predicts cardiovascular outcomes: a systematic review and meta-analysis of prospective studies. Eur Heart J. 2011;32(12):1484–92.
24. Yin J, Jin X, Shan Z et al. Relationship of Sleep Duration With All-Cause Mortality and Cardiovascular Events: A Systematic Review and Dose-Response Meta-Analysis of Prospective Cohort Studies. J Am Heart Assoc. 2017. 6(9):e005947.
25. Hawkins DR, Taub JM, Van de Castle RL. Extended sleep (hypersomnia) in young depressed patients. Am J Psychiatry. 1985;142(8):905–10.
26. Stein PK, Pu Y. Heart rate variability, sleep and sleep disorders. Sleep Med Rev. 2012;16(1):47–66.
27. Marin JM, Carrizo SJ, Vicente E, Agusti AG. Long-term cardiovascular outcomes in men with obstructive sleep apnoea-hypopnoea with or without treatment with continuous positive airway pressure: an observational study. Lancet. 2005;365(9464):1046–53.
28. Koren D, Taveras EM. Association of sleep disturbances with obesity, insulin resistance and the metabolic syndrome. Metabolism. 2018;84:67–75.
29. Morris G, Stubbs B, Köhler CA, et al. The putative role of oxidative stress and inflammation in the pathophysiology of sleep dysfunction across neuropsychiatric disorders: Focus on chronic fatigue syndrome, bipolar disorder and multiple sclerosis. Sleep Med Rev. 2018;41:255–65.
30. Sterne JA, White IR, Carlin JB, et al. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. BMJ. 2009;338:b2393.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.